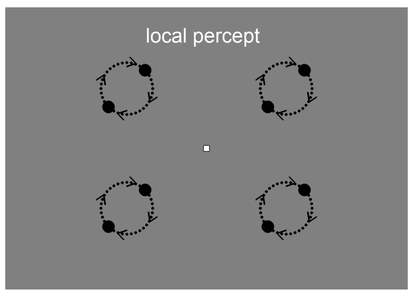
Perceptual grouping reduces perceived speed but does not change uncertainty

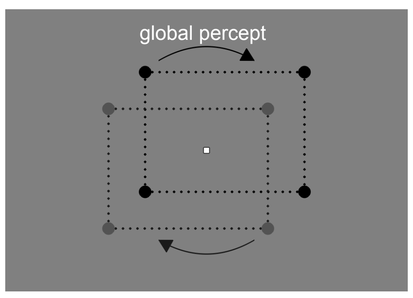
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Background & Motivation

Four dot pairs independently rotating about the vertices of an implied square are perceptually grouped over time, forming the illusion of two overlapping squares[1]. We perceive the global percept to move slower than the local percept, a phenomenon known as the "global slowdown effect".





Prior work has shown that the global slowdown effect cannot be explained by differences in perceived size, emergent rotation, or the number of perceived moving elements between percepts[2].

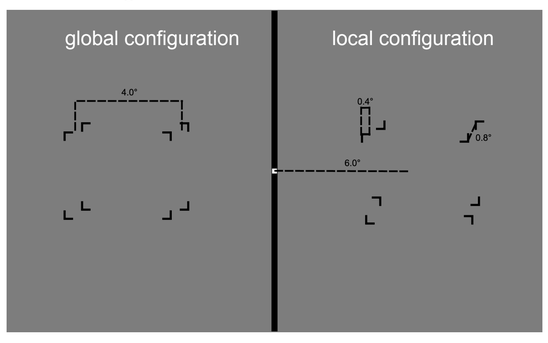
Here, we test 2 additional hypotheses for why the global slowdown effect occurs: Depth cues and Bayesian prior for slow motion.

H1: The effect depends on depth cues introduced by having two elements per vertex of rotation. Removing one element per axis should reduce or eliminate the effect.

H2: The slowdown reflects a systematic difference in sensory uncertainty between global and local percepts. The global percept is more uncertain than the local percept, and a Bayesian prior for slow motion[3] plays a bigger role for global than for local motion.

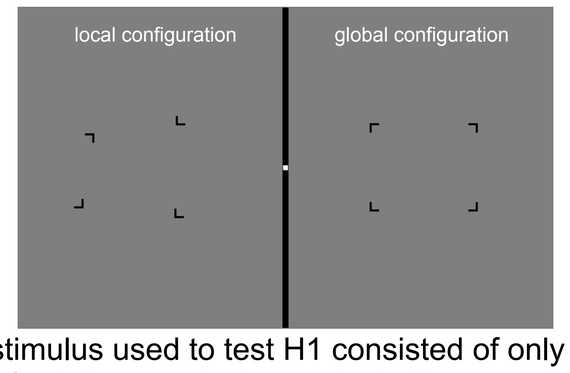
Stimuli

Dots were replaced with "L"-shapes to bias perception towards either global or local motion.



In the global configuration, the Ls were arranged to form the vertices of an illusory square, producing the percept of two overlapping squares moving in phase.

In the local configuration, orientations and phases for Ls along each axis were randomized, making perceptual grouping unlikely.



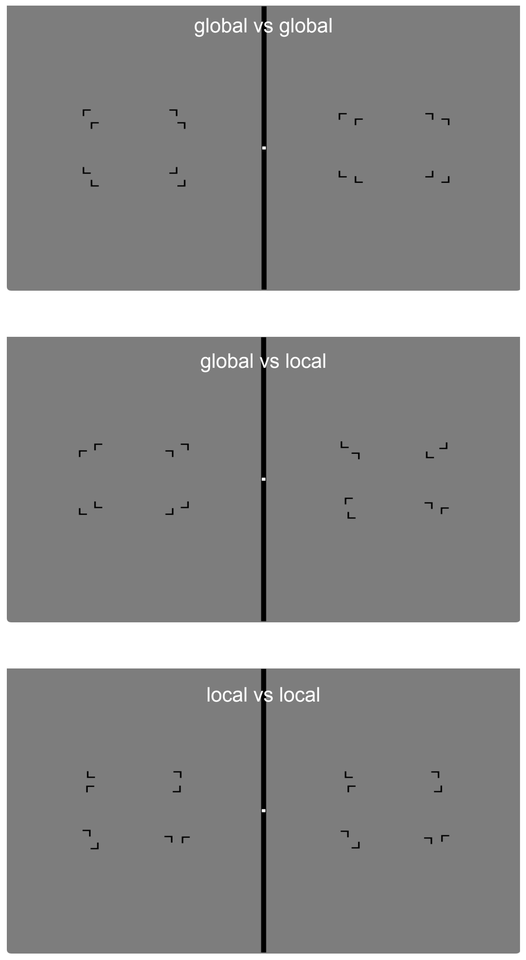
The stimulus used to test H1 consisted of only one L per axis of rotation, producing a single illusory square translating along a circular path.

Methods

2AFC task where participants compared the speed of two configurations on either side of fixation. The reference rotated at 57.6°/s, while the test varied around 57.6°/s.

To test H1, we used two conditions: global vs global and global vs local. To test H2, we introduced local vs local.

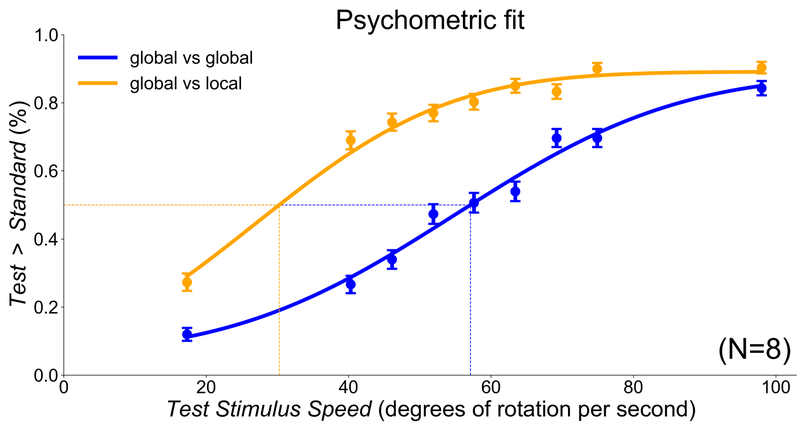
Eye tracking was used to ensure gaze stayed within 2° from fixation each trial. Discarded trials were re-added to the end of the respective block.

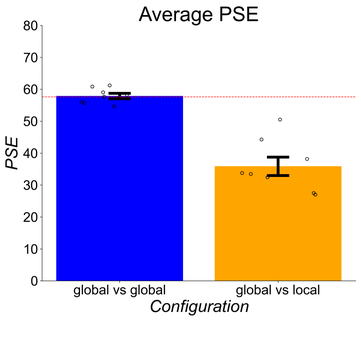


To measure the slowdown effect, we computed the Point of Subjective Equality, focusing on trials with a global reference configuration and a local test configuration.

Sensitivity was measured using the psychometric slope, and uncertainty was assessed by comparing slopes across conditions.

Results: Experiment 1

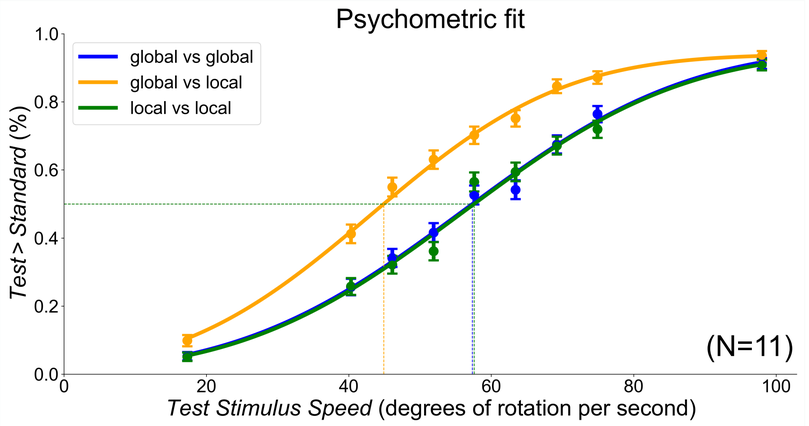


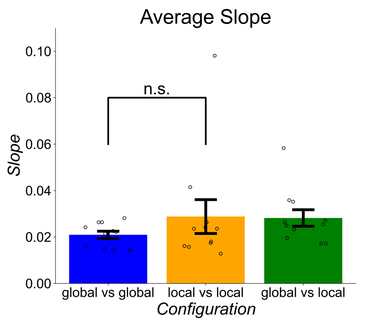


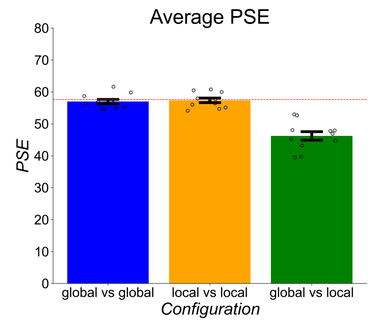
6 participants removed due to poor psychometric function fits ($R^2 < 0.8$ for either condition).

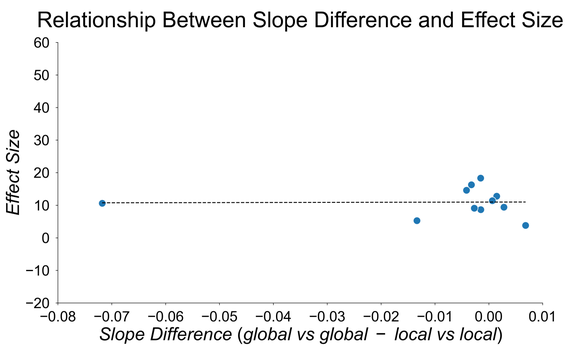
Slowdown effect persisted even with only one L per axis of rotation.

Results: Experiment 2









5 participants removed due to poor psychometric function fits ($R^2 < 0.8$ for any condition).

The slowdown effect was replicated.

No significant difference between the average slope of the global vs global control condition and the local vs local control condition, showing that participants were equally sensitive to changes in speed under both conditions.

No significant relationship between effect size and slope difference. A difference in uncertainty would result in a correlation between slope difference and effect size.

Conclusions

Experiment 1: The global slowdown effect is not caused by depth-like cues that may arise from the grouped percept.

Experiment 2: Global motion is not associated with greater sensory uncertainty.

We find no evidence to support that the slowdown effect is a result of depth-like cues and no evidence to support that the effect is driven by a Bayesian prior for slow motion.

References

[1] Anstis, S. (2003). Levels of motion perception. In L. Harris & M. Jenkin (Eds.), Levels of perception (pp. 75-99). New York: Springer.

[2] Kohler PJ, Caplovitz GP, Tse PU. The global slowdown effect: why does perceptual grouping reduce perceived speed? Atten Percept Psychophys. 2014 Apr;76(3):780-92. doi: 10.3758/s13414-013-0607-x. PMID: 24448695.

[3] Weiss, Y., Simoncelli, E., & Adelson, E. H. (2002). Motion illusions as optimal percepts. Nature Neuroscience, 5(6), 598–604.